

FORGE THE FUTURE HACKATHON · TEAM SANDBOXED

Eyewitness

The incident response agent that saw it happen.

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Elastic · Amazon Bedrock · S3 · Sarvam

The checkout is broken. The dashboard is green. The customer leaves.

Silent Place Order

Handler returns early. No request, no error, no spinner. The shopper clicks three times and gives up.

Backend sees: nothing

Bag total stuck

Quantity goes 1, 2, 3, 4. The line price updates. The order total never moves.

Backend sees: 200 OK

Upload dropped

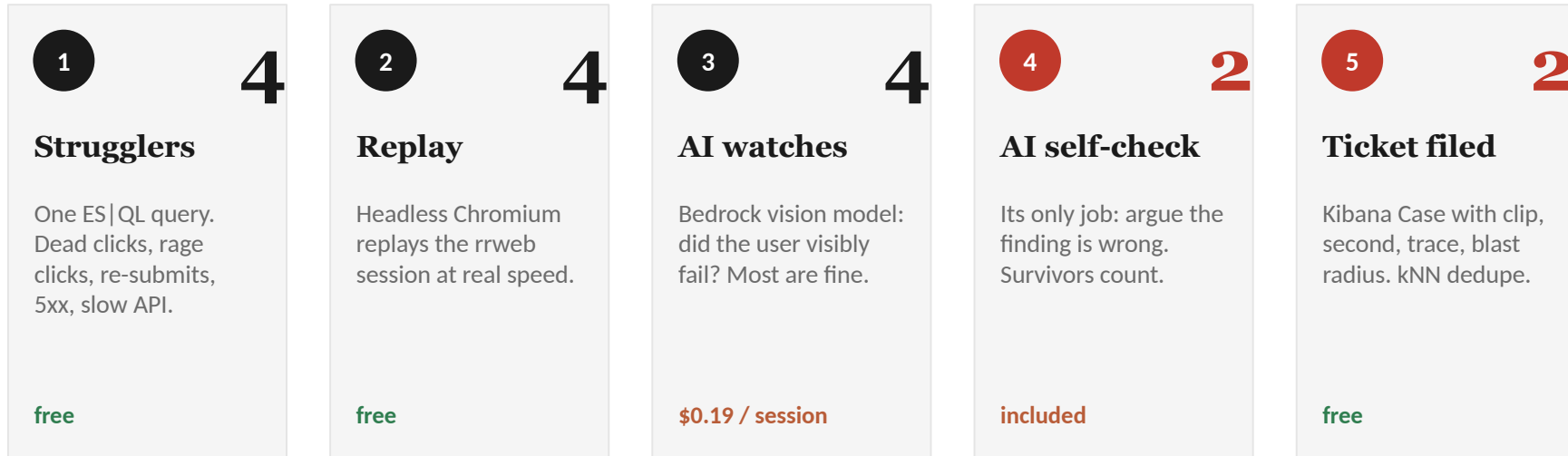
Review photo shows as attached, review posts, the photo is never saved.

Backend sees: 200 OK

Session replay already recorded every one of these, frame by frame. **Nobody watches it.**

Thousands of sessions a day. The first signal is a support ticket, days later, if it comes at all.

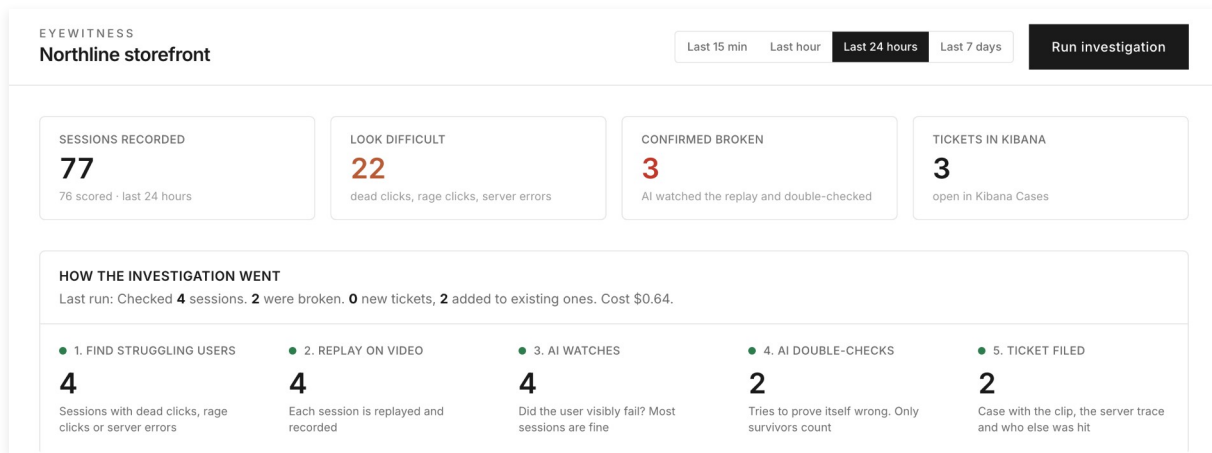
A shrinking funnel: each stage costs more and sees less



Counts are from a real run started from the operator page: 4 sessions flagged, 2 confirmed broken, 2 case updates, 556 seconds, \$0.64.

The model is shown pixels and counts, never a story. It must cite the second.

One button. Watch it work. Read the answer.



The operator page after a run: four numbers, then the pipeline and its result sentence.

Run investigation

Pick a window, press once. No terminal, no config.

Live pipeline

Five stages fill as they run. A plain sentence says what happened.

What is broken

Cards, not reports. Headline, the moment on screen, three steps.

Every session

Journey, signals, score, status and a replay link.

Not “the checkout looks broken.” This.

● OPEN · on /bag · 3 h ago

Order total freezes when you add more items to bag

Seen in this session so far.

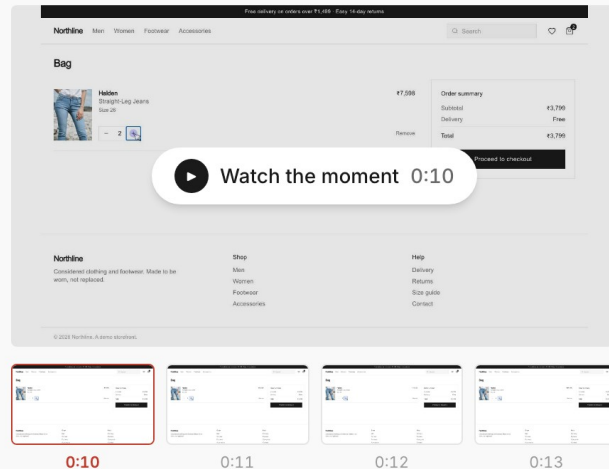
- 1 USER TRIED
Clicked '+' to add more jeans to the bag
- 2 SHOULD HAVE
Order total updates with each extra item
- 3 GOT INSTEAD
Total stayed stuck at ₹3,799 despite 4 items

✓ On screen at 0:10 ✓ Server looked fine (200) ✓ Survived cross-examination · 95%

Open case

Full replay

DETAILS ▼



Plain-English headline

Written by the model for a non-engineer, 60 characters or fewer.

Watch the moment

The clip paused at the failing second. One click plays it.

The three steps

Tried, should have, got instead. The third line is red.

Three checks

Seen on screen. Server fine or failed. Survived review.

Elastic is in the investigation at every stage

FOUR INDICES, ONE SESSION ID

replay-events

raw rrweb event as a disabled object: stored, never indexed

sessions

one summary per session, Painless upsert, signals object

server-events

one doc per API call, session_id + trace_id keywords

incidents

dense_vector 1536 cosine, fingerprint, route, case id

RETRIEVAL, AND WHY

ES|QL the sweep. Threshold and window retunable, no code.

terms + filter aggs backend signals for a thousand sessions in one round trip.

bool + range the request within ± 10 s of the failing second.

kNN incident memory. Cosine gate at 0.85, backend evidence must agree.

Cases API, alerting, Vega the ticket, the pager, the operator dashboard.

AWS Bedrock Converse for triage, extraction, review · Bedrock embeddings · S3 presigned evidence · **Sarvam** Hindi and Tamil case summaries

MEASURED ON THIS BUILD

What we can prove today

100%

precision

0 false positives across 7
healthy sessions

80%

recall

4 of 5 injected faults. The
miss had no signal at all.

7 min

to a ticket

median, session end to
Kibana Case, 3.5 to 16 range

\$0.27

per incident

\$0.19 per session watched.
\$2.13 across 3 runs.

12-session ground-truth run, 5 injected faults, 7 healthy, scored against the truth file. Small and controlled, not a production guarantee.

The number we are most proud of is the zero. Adversarial self-review is what keeps it there.

The value is the gap between “when it broke” and “when someone noticed”

10,000 checkouts a day · 2% silent failure · ₹2,499 average order

	Support-led discovery	Eyewitness, 30-minute sweep
Affected checkouts before discovery	400 to 600 over 2 to 3 days	about 4 per sweep interval
Order value exposed	₹10 to 15 lakh	about ₹10,400 per interval
Engineer time to reproduce	1 to 2 hours hunting for steps	0, scrub to the second
Cost of detection	support headcount, churned customers	about \$6 a day in model calls (USD)

Silent churn

the 96% who never file a ticket become a count

Hours

reproduction goes to zero; the case opens on the second

UI insight

the same signals are a friction report per release

Autonomous QA

a confirmed case is a structured bug report a coding agent can fix

Built to run tonight and to reuse tomorrow

READY

- ✓ **82 tests, lint, types, prod build**
signals, scoring, ingest, admin API, Kibana, memory, renderer
- ✓ **Resumable and scheduled**
per-session checkpoints, scheduled repeats, doctor checks every key first
- ✓ **Cheap by construction**
the sweep discards most sessions before any model call
- ✓ **Honest about limits**
signal-free failures escape; API events, not full OTel

REUSABLE

- ✓ **Three-line integration**
mount the recorder, send the session id header, wrap API routes
- ✓ **Nothing knows about shopping**
generic signals, one query, prompts about a session
- ✓ **Plugs into what you page on**
Kibana Cases, alerting rules, connectors
- ✓ **Same funnel, any journey**
bank onboarding, SaaS settings, insurance claims

The evidence was always there.

Now something watches it, argues with itself, and files only what it can prove.

NEXT

- Close the recall gap: compare the final screen to the expected outcome of the last action
- The fix half of the loop: a coding agent opens the PR with the clip attached
- Fargate renderer, Agent Builder over the indices, APM and RUM, ELSER over runbooks

github.com/someshwaranM/forged-sandboxed · [pull request #1](#) · [SUBMISSION.md](#)

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